CHAPTER IV
METHODOLOGY

This chapter discusses the approaches adopted in addressing the identified problem using a developmental research approach. This serves as the main strategy for the study, focusing on the design, construction, and evaluation of a functional product. With this strategy, the proponents directly address the problem of inefficient manual handling of documents by transforming the theoretical requirements to actual digital repository. This makes sure that the research is not mere observation, but focuses on the technical development and validation of a solution which is of the institutional standards.

Action research has been chosen as the primary methodology since it is an effective way of bridging theory and practice in software development. This will enable the researcher to collaborate with the stakeholders such as librarians and faculty. This partnership guarantees that the eventual solution is technologically satisfactory and that it responds well to the administrative practical needs. Through action research, the proposed study seeks to design, implement, evaluate and enhance the archiving system in close collaboration with end-users, establishing a cycle of constant improvement that can be utilized in the real world.

In addition, the methodology covers critical project management aspects of Gantt chart schedule, project cost estimation, and cost-benefit analysis. The software, system product processes as well as the development and testing procedures are well defined to provide a structured route to project objectives. The strategies of implementation are also outlined and the structured introduction of the software solution to real-life contexts to verify its functionality and utility is mentioned. The chapter provides a solid framework of the system through this comprehensive approach that guarantees the academic rigor and practical applicability in solving the problem identified.

Software Development Methodology

To operationalize the research cycles, the project followed a framework inspired by the Agile Methodology. This framework breaks the project into repeatable iterations, allowing the team to adjust and adapt when unforeseen situations arise during development.

1. Planning Phase

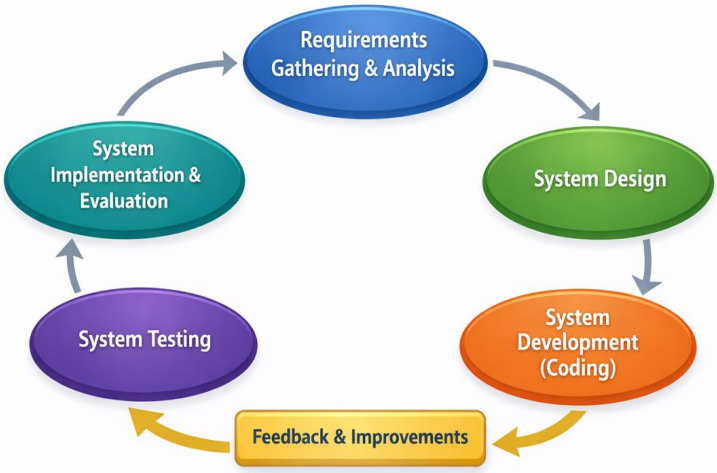


Figure 3.1 *Agile Software Development Methodology*

The project's foundation was established during this phase. We focused on researching institutional needs and defining the core roles within the team to ensure a structured start.

Table 1. Planning Phase Activity

ACTIVITY	DESCRIPTION
Project Initiation Meeting	The team met to brainstorm and delegate potential project ideas that aligned with the capabilities of the team. After evaluating concepts based on feasibility and relevance to university needs, the team selected the Web-based Thesis Archiving System.
Assigning Roles and Responsibilities	A Project Manager was chosen to evaluate each member's skills and capabilities. Based on this assessment, roles were assigned for frontend, backend, documentation, and quality assurance to

	leverage individual strengths.
Benchmarking Key Features	The team researched several related digital repositories and conducted benchmarking to understand their core functionalities. This allowed us to establish the necessary system requirements for our specific development.

Planning Phase Deliverables

- List of project ideas outlining possible features.
- Clear assignment of roles and responsibilities within the project team.
- Benchmarking report describing the key features of related systems.
- Project Plan to officially begin development.

2. System Design Phase

After gathering requirements, the team began drafting the interface and structural designs of the system.

Table 2. System Design Phase Activities

ACTIVITY	DESCRIPTION
Benchmarking Design Concepts	The team, led by the Frontend Developer, reviewed initial concepts for the archiving system to ensure alignment with institutional standards and benchmarked the UI/UX of existing repositories.
Brainstorming for Design	Team members proposed various design concepts, focusing on user-friendly navigation and clear document presentation.
Prototyping	The Frontend Developer created interactive prototypes to visualize the layout, navigation, and user interactions. This provided an actual representation of the system's design before coding began.
Choosing Technological Stack	The analyst and developers researched suitable tools for the project, selecting HTML5 and CSS3 for structural design and styling, JavaScript for client-side interactivity, and PHP for server-side processing and database management.

System Design Phase Deliverables

- Benchmarking Report describing key design concepts.
- High-Fidelity Prototype that reflects the actual look of the archiving system.
- Comparison Table of tools selected for system development.

3. Development Phase

In this phase, the team initiated the construction process, focusing on building the core components and essential functionalities.

Table 3. Development Phase Activities

ACTIVITY	DESCRIPTION
UI/UX Development	The frontend developer transformed the prototype designs into a working website layout using HTML5 and CSS3, ensuring the structure followed the approved mockups and institutional branding.
Development of Essential Features	The backend developer implemented the login features and basic CRUD (Create, Read, Update, Delete) operations for user accounts. We ensured these basic features were stable before adding advanced functions.
Implementation of Key Features	The team collaborated to build the core archiving tools, such as the PDF upload engine and the advanced metadata search filters.
Start of Documentation	Concurrent with development, the team began documenting the technical details and project progress as part of the formal capstone documentation.

Development Phase Deliverables

- Initial Website Layout according to the prototype design.
- Entity Relationship Diagram (ERD) for the data schema.

- Use-Case Diagram outlining system interactions.
- Initial Chapters of the Capstone Documentation

4. Testing Phase

Following development, the team conducted initial testing to verify the system's functionality.

Table 4. Testing Phase Activity

ACTIVITY	DESCRIPTION
Test Planning	The team outlined the objectives for the testing phase, including identifying potential bugs in the search and upload modules.
Functional Testing	The tester compared the system's performance against the Requirements Specification Document to confirm all mandated features were fully operational.

Testing Phase Deliverables

- Testing Plan outlining the objectives and scope of the evaluation.
- Bug/Issue Tracker listing all identified glitches and their resolution status.
- Test Report summarizing the final outcome of the testing phase.

Project Management

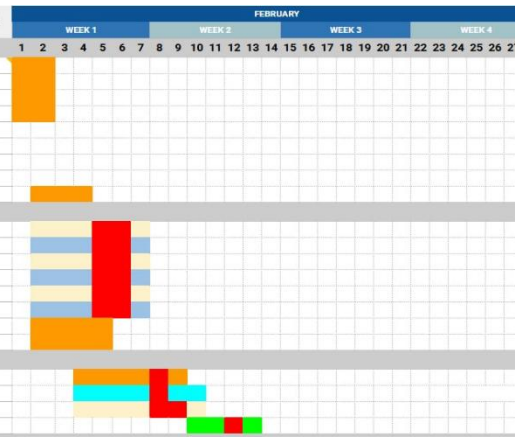
The project was managed in an Agile manner, which enabled the process to remain flexible and at the same time ensured high levels of collaboration during the development. Scheduling was coordinated with the help of tools like Gantt charts, and cost-benefit analysis helped to match the activities and effectively use the available resources. Sprint reviews were consistent, and duties were delegated according to the skills of every member to facilitate accountability and productiveness. This dynamic approach ensured the ability to provide an efficient, user-oriented platform that can address academic and practice requirements.

The project has been made more stable with a proactive risk management strategy implemented early in the development of the project and issues such as integration with the database and security were discussed before they got out of control. Regular meetings enhanced continuous communication, which enabled the development to remain in line with institutional stakeholder needs. Simultaneously, keeping an open line of feedback between the developers and the potential users enabled the refinement of the features of the system continuously. This style of

balance-structured but flexible-helped minimized the delays and had the final repository reliable, well-developed, and in compliance with the archival needs of the university.

Gantt Chart

WBS NUMBER	TASK TITLE	TASK OWNER	NAME	START DATE	DUE DATE
1	**Preliminary Pages**				
1.1	Title Page	Project Manager	Geocallo, Camille Joyce	2/1/26	2/2/26
1.2	Approval Sheet	Project Manager	Geocallo, Camille Joyce	2/1/26	2/2/26
1.3	Acknowledgement	Project Manager	Geocallo, Camille Joyce	2/1/26	2/3/26
1.4	Dedication	Project Manager	Geocallo, Camille Joyce	2/1/26	2/3/26
1.5	Table of Contents	Project Manager	Geocallo, Camille Joyce	2/1/26	
1.6	List of Figures	Project Manager	Geocallo, Camille Joyce	2/1/26	
1.7	List of Tables	Project Manager	Geocallo, Camille Joyce	2/1/26	
1.8	List of Abrreviations / Acronyms	Project Manager	Geocallo, Camille Joyce	2/1/26	2/4/26
1.9	Executive Summary	Project Manager	Geocallo, Camille Joyce	2/2/26	2/4/26
2	**Chapter I - Introduction**				
2.1	Introduction	Documenter	Candilanza, Ivon	2/2/26	2/5/26
2.2	Project Context	Documenter	Pengoc Jorivn	2/2/26	2/5/26
2.3	Purpose and Description	Documenter	Candilanza, Ivon	2/2/26	2/5/26
2.4	Objectives of the system	Documenter	Candilanza, Ivon	2/2/26	2/5/26
2.5	General Objectives	Documenter	Pengoc Jorivn	2/2/26	2/5/26
2.6	Specific Objectives	Documenter	Pengoc Jorivn	2/2/26	2/5/26
2.7	Scope and Limitations	Project Manager	Geocallo, Camille Joyce	2/2/26	2/5/26
2.8	Definition of terms	Project Manager	Geocallo, Camille Joyce	2/2/26	2/5/26
3	**Chapter II - Review of Related Lit and Systems**				
3.1	Related Study 1	Project Manager	Geocallo, Camille Joyce	2/4/26	2/8/26
3.2	Related Study 2	Data Analyst	Dela Victoria Tyrone James	2/4/26	2/8/26
3.3	Related Study 3	Documenter	Candilanza, Ivon	2/4/26	2/8/26
3.4	Synthesis	Developer	Paquit, Mark Kevin Gie	2/10/26	2/12/26
4	**Chapter III - Technical Foundation**				
4.1	Technical Foundation of the System	Developer	Mylene Sellar	3/1/26	3/4/26
4.2	Organizational Chart	Project Manager	Geocallo, Camille Joyce	3/1/26	3/3/26
4.3	Environment	Developer	Paquit, Mark Kevin Gie	3/1/26	3/5/26
4.4	Development Tools and Strategies	Developer	Paquit, Mark Kevin Gie	3/1/26	3/5/26
5	**Chapter IV - Methodology**				
5.1	Methodology	Documenter	Candilanza, Ivon	3/10/26	3/12/26
5.2	Software Development Methodolgy	Project Manager	Geocallo, Camille Joyce	3/11/26	3/15/26
5.3	Project Management	Project Manager	Geocallo, Camille Joyce	3/11/26	3/15/26
5.4	Garitt Chart	Project Manager	Geocallo, Camille Joyce	3/11/26	3/15/26
5.5	Requirements Analysis and Documentation	Documenter	Candilanza, Ivon	3/10/26	3/12/26
5.6	Functional Requirements	Documenter	Candilanza, Ivon	3/10/26	3/12/26
5.7	Non-functional Requirements	Documenter	Candilanza, Ivon	3/10/26	3/12/26
5.8	Design of Software, System Product Processes	Documenter	Candilanza, Ivon	3/10/26	3/12/26
5.9	Entity Relationship Model	Data Analyst	Dela Victoria, Tyrone James	3/9/26	3/12/26
5.1	Use-Case Model	Project Manager	Geocallo, Camille Joyce	3/9/26	3/12/26



Legend:
- Geocallo, Camille Joyce S.
- Sellar, Mylene R.
- Pacquit, Mark Kiven Gie
- Dela Victoria, Tyrone James
- Candilanza, Ivon Lawrence
- Pengoc, Jorvin

Development

WBS NUMBER	TASK TITLE	TASK OWNER	NAME	START DATE	DUE DATE
1	**User & Access Management**				
1.1	User registration	Developer	Mark Kevin Gie Paquit	2/1/26	2/3/26
		Developer	Mylene Sellar	2/3/26	2/6/26
1.2	Design login/logout interface	Developer	Mark Kevin Gie Paquit	2/1/26	2/3/26
		Developer	Mylene Sellar	2/3/26	2/6/26
1.3	Role management	Developer	Mark Kevin Gie Paquit	2/1/26	2/3/26
		Developer	Mylene Sellar	2/3/26	2/6/26
1.4	Session management	Developer	Mark Kevin Gie Paquit	2/1/26	2/6/26
		Developer	Mylene Sellar	2/3/26	2/6/26
1.5	Account activation/deactivation	Developer	Mark Kevin Gie Paquit	2/1/26	2/6/26
		Developer	Mylene Sellar	2/3/26	2/6/26
2	**User Profile Management**				
2.1	Design and implement user profile creation functionality to capture essential user information	Developer	Mark Kevin Gie Paquit	2/1/26	2/4/26
		Developer	Mylene Sellar	2/4/26	2/7/26
2.2	Implement profile viewing and editing functionality to allow users to update their personal information securely	Developer	Mark Kevin Gie Paquit	2/1/26	2/7/26
		Developer	Mylene Sellar	2/1/26	2/7/26
2.3	Develop profile data validation rules to ensure accuracy, completeness, and proper format of user information	Developer	Mark Kevin Gie Paquit	2/1/26	2/7/26
2.3	Implement profile photo upload functionality with file type and size restrictions, and storage in the system database	Developer	Mark Kevin Gie Paquit	2/1/26	2/7/26

3	**Student Project List**				
3.1	Design and implement a project listing interface to display all submitted thesis documents with sorting and filtering options	Developer	Mark Kevin Gie Paquit	2/7/26	2/11/26
		Developer	Mylene Sellar	2/11/26	2/15/26
3.3	Implement a project details view to display complete thesis information including metadata, uploaded file, and current status	Developer	Mark Kevin Gie Paquit	2/7/26	2/11/26
		Developer	Mylene Sellar	2/11/26	2/15/26
3.4	Implement role-based edit and delete functionality for thesis records, allowing only authorized users to modify or remove submissions	Developer	Mark Kevin Gie Paquit	2/7/26	2/11/26
		Developer	Mylene Sellar	2/11/26	2/15/26
3.5	Develop a project status tracking feature to display current thesis status	Developer	Mark Kevin Gie Paquit	2/7/26	2/11/26
		Developer	Mylene Sellar	2/11/26	2/15/26
4	**Student Thesis Submission Module**				
4.1	Design and implement a thesis file upload interface for students with user-friendly input controls	Developer	Mark Kevin Gie Paquit	2/17/26	2/20/26
4.2	Implement file validation to enforce allowed formats (e.g., PDF, DOCX) and maximum file size limits during upload	Developer	Mylene Sellar	2/20/26	2/22/26
4.3	Develop functionality to automatically record submission timestamps for each uploaded thesis	Developer	Mark Kevin Gie Paquit	2/17/26	2/20/26
4.4	Implement a revision upload feature to allow students to resubmit updated versions of their thesis	Developer	Mark Kevin Gie Paquit	2/17/26	2/20/26
4.5	Develop a submission history tracking system to record all uploads, including version, date, and status	Developer	Mark Kevin Gie Paquit	2/17/26	2/20/26
4.6	Implement automatic linking of uploaded thesis files to their corresponding metadata records in the database	Developer	Mark Kevin Gie Paquit	2/17/26	2/20/26

5	**Thesis Review and Approval Workflow**				
5.1	Design and implement a thesis review interface for research advisers with functionality to add comments and provide feedback on submissions	Developer	Mark Kevin Gie Paquit	2/20/26	2/23/26
5.2	Implement approval and rejection functionality, including the ability to return theses for revision with specified remarks	Developer	Mylene Sellar	2/23/26	2/26/26
5.3	Develop a review history tracking system with timestamps, reviewer identity, and action taken	Developer	Mark Kevin Gie Paquit	2/20/26	2/23/26
5.4	Implement automated notifications to inform students and relevant users of review outcomes	Developer	Mylene Sellar	2/23/26	2/26/26
5.5	Design a coordinator dashboard to monitor and manage pending thesis approvals across departments	Developer	Mark Kevin Gie Paquit	2/20/26	2/23/26
5.6	Implement role-based workflow control to enforce departmental hierarchy	Developer	Mylene Sellar	2/23/26	2/26/26
6	**Metadata Management**				
6.1	Design and implement metadata input forms including fields such as title, author(s), abstract, keywords, department, and publication year	Developer	Mark Kevin Gie Paquit	3/1/26	3/4/26
		Developer	Mylene Sellar	3/5/26	3/7/26
6.2	Implement metadata validation rules to ensure completeness, correct format, and required field input before submission	Developer	Mark Kevin Gie Paquit	3/1/26	3/4/26
		Developer	Mylene Sellar	3/5/26	3/7/26
6.3	Develop functionality to store metadata and link it to the corresponding thesis file in the database	Developer	Mark Kevin Gie Paquit	3/4/26	3/4/26
		Developer	Mylene Sellar	3/5/26	3/7/26
6.4	Implement an edit feature to allow authorized users to update existing metadata records	Developer	Mark Kevin Gie Paquit	3/4/26	3/4/26
		Developer	Mylene Sellar	3/4/26	3/5/26
6.5	Integrate metadata with search and filtering functions to enable efficient retrieval of thesis documents	Developer	Mark Kevin Gie Paquit	3/1/26	3/4/26
		Developer	Mylene Sellar	3/5/26	3/7/26

7	**Thesis Archive and Repository Management**				
7.1	Design and implement structured repository folders categorized by department and academic year	Developer	Mark Kevin Gie Paquit	3/8/26	3/11/26
		Developer	Mylene Sellar	3/12/26	3/15/26
7.2	Implement validation to store only approved thesis documents in the archive	Developer	Mark Kevin Gie Paquit	3/8/26	3/11/26
		Developer	Mylene Sellar	3/12/26	3/15/26
7.3	Develop a system to record and track thesis storage logs (upload date, uploader, file details)	Developer	Mark Kevin Gie Paquit	3/8/26	3/11/26
		Developer	Mylene Sellar	3/12/26	3/15/26
7.4	Implement secure download and view access for authorized users	Developer	Mark Kevin Gie Paquit	3/8/26	3/11/26
		Developer	Mylene Sellar	3/12/26	3/15/26
7.5	Develop an internal browsing feature with search and filter options for thesis reference	Developer	Mark Kevin Gie Paquit	3/8/26	3/11/26
		Developer	Mylene Sellar	3/12/26	3/15/26
8	**Notification and Status Tracking Module**				
8.1	Implement student thesis submission status tracking with automated submission alerts	Developer	Mark Kevin Gie Paquit	2/1/26	2/5/26
		Developer	Mylene Sellar	2/5/26	2/10/26
8.2	Develop an automated archival notification system to inform users when a thesis is approved and archived	Developer	Mark Kevin Gie Paquit	2/1/26	2/5/26
8.3	Design and implement a role-based dashboard notification system for students, faculty, and administrators	Developer	Mylene Sellar	2/5/26	2/9/29
8.4	Develop a notification history log to record all sent notifications	Developer	Mylene Sellar	2/5/26	2/9/29
9	**Dashboard and Monitoring Module**				
9.1	Design and implement a student dashboard displaying submission status, notifications, and thesis progress tracking	Developer	Mark Kevin Gie Paquit	2/5/26	2/7/26
9.2	Develop a research adviser dashboard for reviewing submissions, providing feedback, and tracking assigned students	Developer	Mylene Sellar	2/7/26	2/9/26
9.3	Implement a research coordinator dashboard to monitor overall submissions, approval status, and departmental reports	Developer	Mark Kevin Gie Paquit	2/5/26	2/7/26
9.4	Design a department dean dashboard for viewing summarized thesis reports, approval status, and performance metrics	Developer	Mylene Sellar	2/7/26	2/9/26
9.5	Develop a Librarian dashboard for final archiving of approved theses, managing metadata and access permissions, tracking archived records, and monitoring repository usage statistics.	Developer	Mylene Sellar	2/7/26	2/9/26
9.6	Develop an admin dashboard for managing users, system activity logs, and overall system configuration	Developer	Mark Kevin Gie Paquit	2/5/26	2/7/26
9.7	Implement data visualization features (graphs and charts) to display thesis submission trends, approval rates, and archive statistics	Developer	Mylene Sellar	2/7/26	2/9/26
9.8	esign and integrate quick navigation links for accessing frequently used actions (e.g., submit thesis, review documents, generate reports)	Developer	Mark Kevin Gie Paquit	2/5/26	2/7/26

10	**Version Control Module**				
10.1	Implement a version tracking system to record and list all published versions of each thesis	Developer	Mark Kevin Gie Paquit	3/17/26	3/21/26
10.2	Develop functionality to store version details, including version number, publication date, and author/uploader	Developer	Mylene Sellar	3/22/26	3/24/26
10.3	Design a version history interface to allow users to view and compare all published versions of a thesis	Developer	Mark Kevin Gie Paquit	3/17/26	3/21/26
10.4	Implement a feature to identify and highlight the latest version of each thesis document	Developer	Mylene Sellar	3/22/26	3/24/26

Other Development

WBS NUMBER	TASK TITLE	TASK OWNER	NAME	START DATE	DUE DATE
1	**Database Development**				
2.1	Design database schema	Data Analyst	Dela Victoria, Tyrone James	2/10/26	2/12/26
2.2	Create database tables and relationships	Data Analyst	Dela Victoria, Tyrone James	2/12/26	2/14/26
2.3	Implement data constraints and indexing	Data Analyst	Dela Victoria, Tyrone James	3/1/26	3/2/26
2.4	Set up database security (encryption, access roles)	Data Analyst	Dela Victoria, Tyrone James	3/4/26	3/6/26

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Project Cost Estimate

Hardware Cost

ITEM	DESCRIPTION	QTY	UNIT COST	ANNUAL COST
Acer Intel/ Aspire 5	Laptop	2	44,999	88,998
PC/ Tuf	Computer	1	50,999	50,999
Mouse+Keyboard		2	1,500	3,000

Software Cost

Git &GitHub	Version Control Team	Team(Subscription)	₱222.00	₱2,664.00

VS Code	Code Editor	Open-Source	₱0.00	₱0.00
Figma	Design Prototyping Tool	Professional Collab Set(Subscription)	₱0.00	₱0.00
Tailwind CSS	framework	Open-Source	₱0.00	₱0.00
Three JS	Front-end Library for 3D Designs	Open-Source	₱0.00	₱0.00

Supply Cost

Item	Description	Unit Cost	Annual Cost
Internet	Used for research and development	₱1,100.00	₱14,300.00
ElectricityBill	Used for keeping the devices working	₱1,100.00	₱12,000,00

Cost Type	Total Cost
Hardware Cost	₱142,997.00
Software Cost	₱2,664.00
Supply Cost	₱26,300.00
Overall Total Cost	₱171,91

Cost and Benefit Analysis

Year	FV	(1+i)n	PV	Accumulated Value
1	PHP 59,860	1.10	PHP 65,846	PHP 65,846
2	PHP 59,860	1.21	PHP 72,430.60	PHP 138,276.60
3	PHP 59,860	1.331	PHP 79,673.66	PHP 217,950.26

Table No. N: Payback Period

The table provides a structured approach to estimating the future benefits (FV) of the capstone project over a three-year period. It calculates the present value (PV) of these future benefits using a discount rate of 10% annually. The accumulated value shows the total estimated benefit over time, accounting for the time value of money and the impact of compounding interest.

This analysis helps justify investments in terms of both financial and non-financial outcomes. It enables project managers and stakeholders to identify potential risks, assess alternative solutions, and align project goals with organizational or academic objectives. Furthermore, the analysis provides a framework for comparing different project scenarios or investment options, facilitating the selection of the most economically sound and strategically beneficial course of action. Ultimately, by quantifying and qualifying the costs and benefits associated with the project, a comprehensive cost and benefit analysis ensures transparency, accountability, and informed decision-making throughout the project lifecycle.

Requirements Analysis and Documentation

The Requirements Analysis and Documentation phase is the most critical part of analysis between the identified issues in the institutions and the technical solution. The phase is based on the developmental research approach, targeting to put the needs of the research department of the university into a definite set of specifications. Through careful recording of these requirements, the proponents form a sort of contractual agreement of functionality that guarantees that the ultimate Web-based Thesis Archiving System would deal with these inefficiencies of manual document handling. This step will avoid scope creep and will also give the evaluation criteria that it will apply in the subsequent testing phases of the project.

Functional Requirements

CODE	USE CASES	DESCRIPTION	ACTORS
FR-01	Login / Authentication	Authenticate users via institutional credentials with role-based access.	All Actors
FR-02	Manage Profile	Update personal details, security questions, and notification preferences.	All Actors
FR-03	Submit Thesis	Upload PDF documents and input mandatory metadata.	Student
FR-04	Version Control System	Track document revisions and allow retrieval of previous drafts.	Student, Adviser
FR-05	Thesis Review	Faculty evaluation of drafts through comments and status updates.	Adviser, Coord., Dean
FR-06	Monitor Approval Workflow	Visual tracking of the current stage in the multi-level approval process.	Student, Coord.
FR-07	Receive Notification	Automate email or system alerts for status changes or feedback.	Student, Faculty
FR-08	View Project Lis	Consolidated view of all active research projects and their statuses.	Coordinat or, Dean
FR-09	Browse Repositories	Searchable gallery for public or institutional access to archived works.	Librarian, Visitors
FR-10	Archive Thesis	Move verified research into a permanent, read-only digital vault.	Librarian
FR-11	View Dashboards	Visual reporting on submission trends and archival growth statistics.	Admin, Dean

FR-12	Manage Users	Create, suspend, or modify accounts and assign institutional roles.	Admin
FR-13	View Audit Logs	Maintain a chronological record of all system-wide transactions.	Admin
FR-14	Backup/Restore	Schedule automated data redundancy to prevent repository loss.	Admin
FR-15	Logout	Securely terminate active sessions and clear temporary cache.	All Actors

Non-functional Requirements

Performance

- The system is designed to support a minimum of 100 users concurrently without experiencing much degradation in performance and maintain stability during peak periods in the academic calendar.

- Normal load Response time to frequent operations (search queries, metadata) shall be less than 2 seconds.

Security

- All the data transfer is secured with the help of the HTTPS protocol to ensure that the information is encrypted and no unauthorized access is allowed.

- The privileges are restricted and role-based controls are applied to make sure that the privacy of data in the administrative and review modules is guaranteed and to avoid any unauthorized activities.

Usability

- The user interface is made easy to use and completely responsive, offering uniformity to desktop, tablet, and mobile-based systems.

- The development is based on standards of accessibility (e.g., WCAG) so that the repository could be used by people with disabilities and inclusivity within the university would be encouraged.

Reliability

- The expected minimum meaning of uptime in the system is 99.9% because the system is to be made available to students and faculty during the normal business and library hours.

- The process of extensive backup and recovery has been designed to reduce possible data loss, and quick restoration is achieved in the event of system failure.

Scalability

- The system has been designed with the option of scaling to meet the requirements of future increase in the number of users and total amount of archived research data.
- Database and server availability is optimally configured to support horizontal and vertical scalability, enabling the platform to grow with the increase of digital collection of the university.

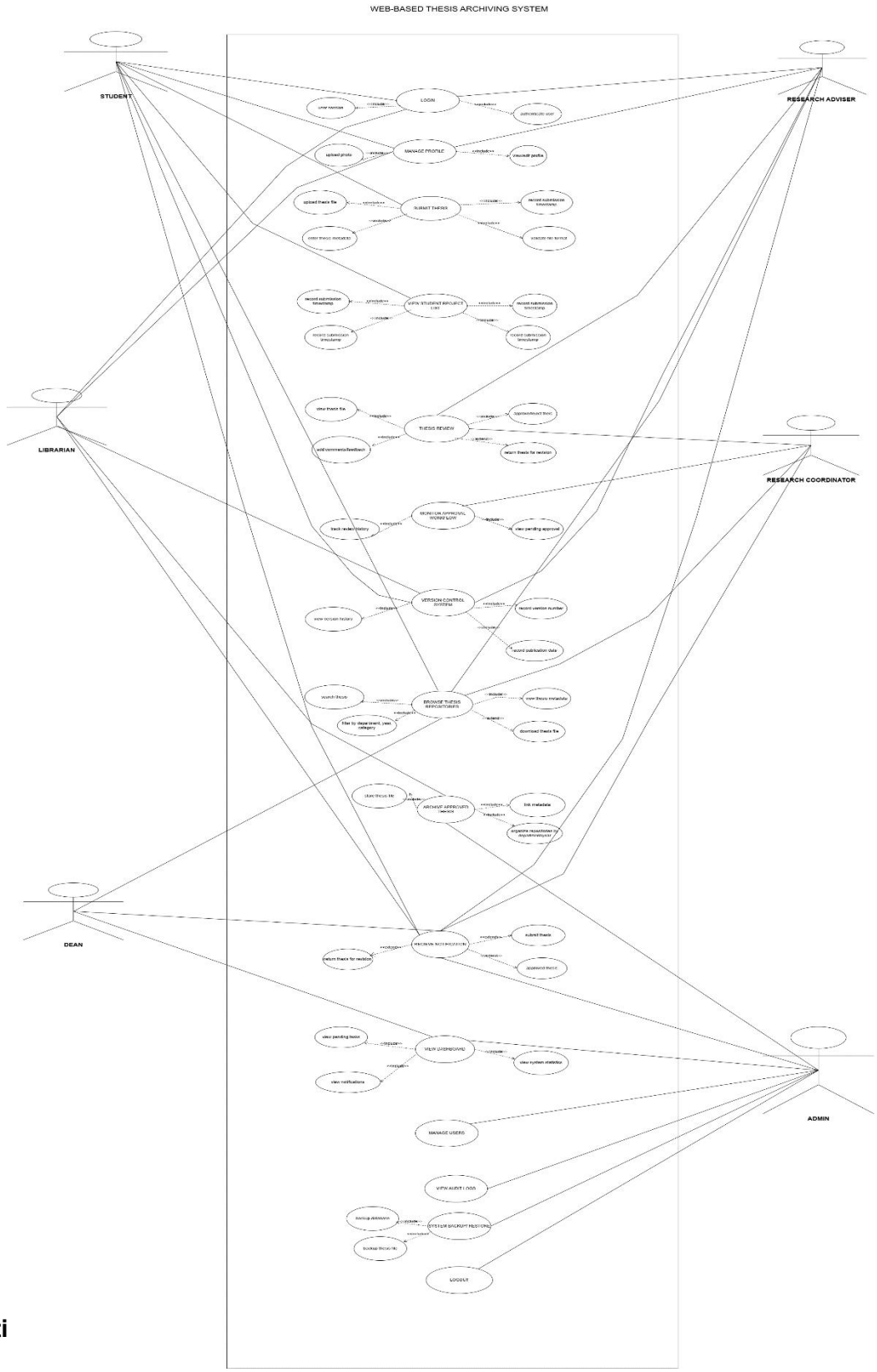
Design of Software, System Product Processes

Design of Software and System Product Processes is the phase to provide a transition structure between conceptual requirements and technical blueprint. This phase involves determining the internal logic, data flow and user interaction models of the Web-based Thesis Archiving System. The proponents make sure that the system is not only functional but also scalable and maintainable, by creating a clear architectural design. The design process helps mitigates the disconnect between what the system has to do and how it can technically accomplish those objectives and it gives a visual and logical map over which the development team can apply.

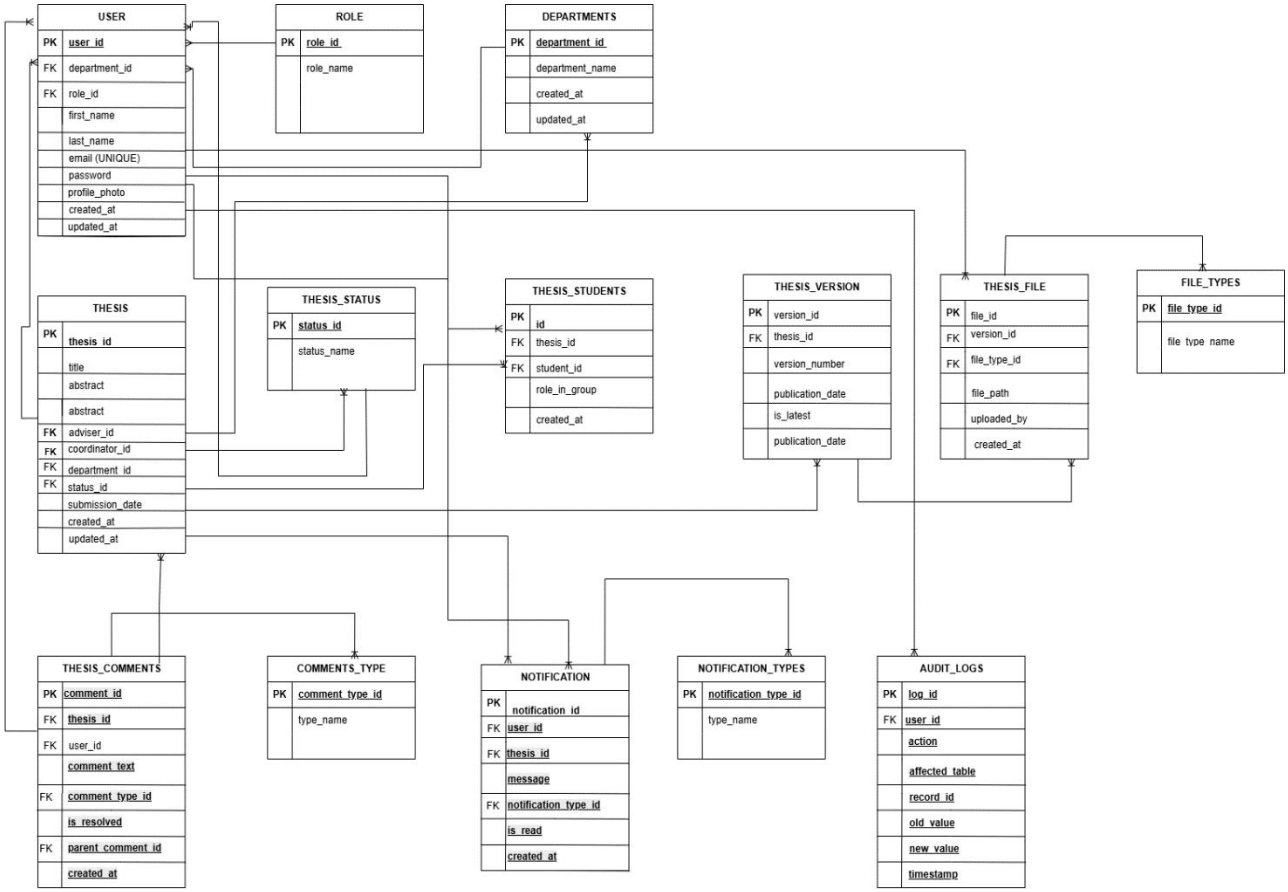
Some of the critical layers included in software design are the User Interface design (UI), database schema, and backend logic of the software. In this case, the design has focused on an autonomy of workflow of the six main actors in this project; this includes the process of submission by the student up to the administration approval processes. Strikingly, system product processes are strictly mapped to show how data crosses the platform, and every passage is secure, and properly documented, e.g. a thesis progressing through the draft, to an approved archive.

Moreover, this section outlines the combination of the selected technology stack, i.e. how React.js will synchronize with the presentation layer whereas Python and Mongo Data will complement the powerful needs of the data process and storage. The project also reduces technical debt and logic errors during the coding phase by definition of these processes upfront. Finally, the software design and product processes enable the end product to be a powerful, well-performing tool with the nuances to maintain the intellectual legacy of the university with academic integrity.

Use-Case Model


WEB-BASED THESIS ARCHIVING SYSTEM
STUDENT
RESEARCH ADVISER
LIBRARIAN
RESEARCH COORDINATOR
DEAN
ADMIN
LOGIN
enter credentials
authenticate user
MANAGE PROFILE
upload photo
visualize profile
SUBMIT THESIS
upload thesis file
enter thesis metadata
record submission timestamp
validate file format
VIEW STUDENT PROJECT LIST
record submission timestamp
record submission timestamp
record submission timestamp
THESIS REVIEW
view thesis file
add comments/feedback
approve/reject thesis
return thesis for revision
MONITOR APPROVAL WORKFLOW
track review history
view pending approval
VERSION CONTROL SYSTEM
view version history
record version number
record publication date
BROWSE THESIS REPOSITORIES
search thesis
filter by department, year, category
view thesis metadata
download thesis file
ARCHIVE APPROVED THESIS
store thesis file
link metadata
organize repositories by department/year
RECEIVE NOTIFICATION
return thesis for revision
submit thesis
approved thesis
VIEW DASHBOARD
view pending tasks
view notifications
view system statistics
MANAGE USERS
VIEW AUDIT LOGS
SYSTEM BACKUP RESTORE
backup database
backup thesis file
LOGOUT

Enti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FK role_id
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last_name
email (UNIQUE)
password
profile_photo
created_at
updated_at
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role_name
DEPARTMENTS
PK department_id
department_name
created_at
updated_at
THESIS
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abstract
abstract
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FK coordinator_id
FK department_id
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is_read
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AUDIT_LOGS
PK log_id
FK user_id
action
affected_table
record_id
old_value
new_value
timestamp